Exam

Instructions

- The exam will be graded out of 81 points. You need 35 points to pass The grade of 1.0 is given for 65 points or more. This allows you to choose which questions you want to answer.
- The total duration of the exam is 2 hours.
- The exam is divided into two sections: *direct application exercises*, which are straightforward questions based directly on the course material, and *problems*, which are more complex and open-ended.
- We do not expect you to fully solve the problems; partial answers can still earn points. We look for your ability to tackle unfamiliar challenges, propose coherent ideas, and discuss their limitations. However, answers without proper justification will not be awarded points.
- Please ensure your answers are clear and readable. Draft your ideas before writing them on the exam sheet. While minor corrections are acceptable, your answers must remain legible.
- The course content does not cover machine learning, except for Bayesian networks. Therefore, solutions involving other machine learning techniques (e.g., LLMs, deep learning algorithms, or reinforcement learning) will not be accepted.
- You have space to write your answers directly below each question. If you require additional space, use the blank sheets provided at the end and clearly reference the question number.
- Please avoid using red ink.
- Write your name and matriculation number on top of each page.

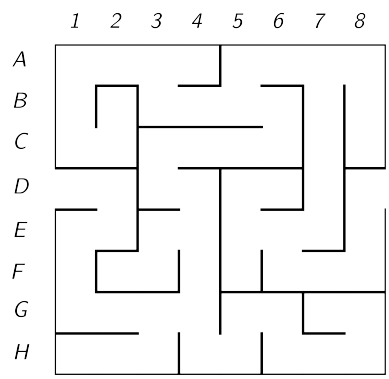
1 Direct application exercises (35 points)

1.1 Search in trees (3 points)

Question 1 (3 points)**.** Prove that the BFS strategy finds the shallowest solution, for any problem with a finite action space.

1.2 Partial observability (5 points)

Question 2 (3 points)**.** We consider an agent in a maze with partial observability: the agent can only see the walls in its current cell and knows its orientation (i.e. it knows exactly in which sides the walls are). The agent knows the map of the labyrinth, but ignores where it is located. It starts from a random cell, which is unknown to it.



1. In its initial cell, the agent can see a wall above and below, and no wall on the left and right. What is its belief state?

2. Of actions `Left`, `Right`, `Up` and `Down`, which one should it play to filter its belief space maximally?

Question 3 (2 points)**.** Explain the A^* algorithm and the notion of heuristic.

1.3 Game theory (8 points)

In the following questions, Player 1 plays the rows and Player 2 the columns.

Question 4 (2 points)**.** What are the pure Nash equilibria of the following bi-matrix game?

$$G = \begin{pmatrix} 1,3 & 5,2 & 4,3 & 1,2 & 6,2 \\ 2,6 & 4,-1 & 3,4 & 1,1 & 2,3 \\ 4,6 & 0,1 & 5,3 & 2,1 & 4,4 \end{pmatrix}$$

Question 5 (3 points)**.** What are the mixed Nash equilibria of the following bi-matrix game?

$$G = \begin{pmatrix} 6,4 & 2,0 \\ 2,-2 & 4,2 \end{pmatrix}$$

Question 6 (3 points)**.** Are these two statements correct? If so, give an example of corresponding 2-player games. Otherwise, justify your answer.

1. There exist games which do not possess any Nash equilibrium in **pure** strategy.

2. There exist games which do not possess any Nash equilibrium in **mixed** strategy.

1.4 Logic (16 points)

Question 7 (2 points)**.** If a, b and c are propositional variables, is the following sentence a tautology? Unsatisfiable? Justify your answer.

$$(a \wedge (c \rightarrow b)) \rightarrow (\neg c \leftrightarrow (a \rightarrow b))$$

Question 8 (4 points)**.** If ϕ and ψ are two propositional formulae, prove that

$$\phi \wedge \psi \quad \equiv \quad \psi \wedge \phi$$

Question 9 (3 points)**.** Which of the following sentences are syntactically valid first-order logic sentences (the comma is in principle not in the syntax of FOL, but used here to make formulae easier to read):

- $\forall x,\ f(a) \to \neg f(\neg a)$
- $\forall x,\ \forall f,\ f(f(x)) = x$
- $\exists b,\ R(a, b)$

Question 10 (2 points)**.** In description logics, we consider the following interpretation $\mathcal{I}$:

- $\mathcal{D}^{\mathcal{I}} = \{e_1, e_2, e_3, e_4, e_5, e_6\}$
- Concepts: $A^{\mathcal{I}} = \{e_1, e_2, e_3\}$, $B^{\mathcal{I}} = \{e_3, e_5\}$, $C^{\mathcal{I}} = \{e_2, e_4, e_6\}$, and $D^{\mathcal{I}} = \{e_1, e_3, e_5\}$
- Roles: $r^{\mathcal{I}} = \{(e_2, e_4), (e_3, e_5), (e_4, e_1)\}$, $s^{\mathcal{I}} = \{(e_6, e_2), (e_4, e_5), (e_3, e_3)\}$, and $t^{\mathcal{I}} = \{(e_3, e_4), (e_4, e_5), (e_6, e_6)\}$

For each of the following concepts $\mathcal{C}$, what is the corresponding set $\mathcal{C}^{\mathcal{I}}$?

1. $A \sqcap C$
2. $\exists r.D$
3. $\forall s.B$
4. $\exists t.\forall r.(A \sqcup B)$

Question 11 (3 points)**.** In deontic logic, the modal operator $\Box$ reads as obligation (i.e. $\Box\phi$ reads "ϕ is obligatory". Based on this interpretation, how do you read the following properties, and do you think they should be satisfied in deontic logic:

1. $\Box(\phi \to \psi) \to (\Box\phi \to \Box\psi)$:
2. $\Box\phi \to \phi$
3. $\Box\phi \to \Diamond\phi$

Question 12 (2 points)**.** Decompose the following knowledge base KB in the description logic $\mathcal{ALC}$ in its ABox and TBox.

$$\begin{aligned} KB = \{ \quad & \text{Heroine} \equiv \text{Hero} \sqcap \text{Female}, \\ & \text{Hero(Batman)}, \\ & \text{Female}(Wonderwoman), \\ & Criminal \sqsubseteq \exists \text{fight.Hero} \quad \} \end{aligned}$$

1.5 Natural Language Processing (3 points)

Question 13 (3 points)**.** Consider the following context-free grammar, with S' as the start symbol:

$$
\begin{array}{lcl}
S & \rightarrow & A \mid \varepsilon \\
A & \rightarrow & T \mid (N, C) \\
C & \rightarrow & C, A \mid A \\
T & \rightarrow & a \mid b \mid c \\
N & \rightarrow & x \mid y \mid z
\end{array}
$$

Which of the following strings are true in the language generated by this grammar:

1. `x`
2. `c`
3. `(z, a, b, a, b, c)`
4. `(x, (y, a), (z, b)`
5. `(x, (x, (x, a)))`
6. `(x, a, (y, b), c)`

2 Problems (46 points)

2.1 Kripke's semantic (10 points)

We remind that, in modal logic, a **Kripke relational frame** is a pair $\mathcal{F} = \langle W, R \rangle$, where W is a non-empty set of worlds and $R \subseteq W \times W$ is the accessibility relation. When $(v, w) \in R$, we say that w is a successor of v, which is written vRw. A **Kripke model** based on the frame $\mathcal{F} = \langle W, R \rangle$ is a triple $\mathcal{I} = \langle W, R, \pi \rangle$, with π a function mapping worlds to a truth assignment. We write $\mathcal{I}, w \models \phi$ to express the fact that a formula ϕ is true in world w in interpretation $\mathcal{I}$. A formula ϕ is true, if it is true for all $\mathcal{I}$ and w.

Question 14 (2 points)**.** A desired property is that: a formula that does not contain modal operators $\Box$ and $\Diamond$ is true in $\mathcal{I}, w$ if it is true as a propositional formula in world w with propositional interpretation π_w. For instance:

$$\mathcal{I}, w \models a \qquad \text{iff} \qquad \pi_w(a) = True.$$

In a similar way, what is a necessary and sufficient condition for $\mathcal{I}, w \models \phi \rightarrow \psi$?

The semantic of the modal operator $\Box$ is defined by $\mathcal{I}, w \models \Box\phi$ iff $\mathcal{I}, v \models \phi$ for all v such that wRv.

Question 15 (4 points)**.** Assume that interpretation $\mathcal{I}$ is such that wRw for all world w. Show that $\mathcal{I}, w \models \Box\phi \rightarrow \phi$

Question 16 (4 points)**.** Assume that interpretation $\mathcal{I}$ is such that if wRv and vRu, than wRu for all worlds u, v and w. Show that $\mathcal{I}, w \models \Box\phi \rightarrow \Box\Box\phi$.

2.2 Rock-paper-scissor (8 points)

Question 17 (8 points)**.** Show that the mixed strategies $p = (1/3, 1/3, 1/3)$ for Player 1 and $q = (1/3, 1/3, 1/3)$ form a Nash equilibrium (p, q) for the Rock-Paper-Scissor game:

P1\P2	Rock	Paper	Scissors
Rock	$(0, 0)$	$(-1, 1)$	$(1, -1)$
Paper	$(1, -1)$	$(0, 0)$	$(-1, 1)$
Scissors	$(-1, 1)$	$(1, -1)$	$(0, 0)$

2.3 Edit distance (8 points)

An edit distance is a measure of how many operations are needed to transform one character string x into one character string y. For instance, the distance between `play` and `plays` is expected to be shorter than the distance between `play` and `played`, since it implies adding only one letter instead of two.

Question 18 (8 points)**.** Propose a definition of an edit distance. For this exercise, you only require from this distance that $d(x, x) = 0$. In particular, you do not require that $d(x, y) = d(y, x)$. To define the distance, propose and formalize some elementary transformations on strings (for instance adding character), and describe an algorithm to compute $d(x, y)$.

2.4 Context-free grammar for instructions to a robot (10 points)

The goal of this exercise is to design a grammar for a simple subset of English arithmetic commands used by a robotic arm. The language includes:

1. Actions: `pick`, `drop`, `move`
2. Objects: `box`, `ball`, `pyramid`
3. Optional qualifiers: `red`, `blue`, `big`
4. Locations: `left`, `right`, `center`
5. Operators: `and`
6. Optional modifiers: `quickly`, `slowly`

Valid commands should be for instance "pick box", "move blue ball to left" and "pick ball quickly and move big box to center slowly".

Question 19 (10 points)**.** Propose a context-free grammar that can generate all valid commands described above. In particular, the grammar must handle optional modifiers after an action or an object, optional qualifier before and object and the conjunction of multiple commands.

2.5 Towards a semantics of fuzzy logic (10 points)

In fuzzy logic, we associate propositions ϕ to truth degrees $\mu(\phi) \in [0, 1]$, with the convention that the higher the degree, the "more true" the proposition. In particular $\mu(\phi) = 0$ means that ϕ is false and $\mu(\phi) = 1$ means that ϕ is true. To define the semantic of fuzzy logic, we then need to evaluate for $\mu(a \wedge b)$, $\mu(a \vee b)$ and $\mu(\neg a)$. We express these with functions $c : [0, 1] \to [0, 1]$, $s : [0, 1]^2 \to [0, 1]$ and $t : [0, 1]^2 \to [0, 1]$ such that:

- $\mu(\neg a) = c(\mu(a))$
- $\mu(a \vee b) = s(\mu(a), \mu(b))$
- $\mu(a \wedge b) = t(\mu(a), \mu(b))$

Question 20 (8 points)**.** Discuss the mathematical properties that the functions c, s and t should satisfy to be consistent with the desired properties of negation, conjunction and disjunction (*Hint: think of their properties in propositional logics*). Propose possible choices for such functions.

Question 21 (2 points)**.** Using the functions c, s and t, express how you would define $\mu(a \to b)$ and $\mu(a \leftrightarrow b)$.